

		Codes	IC1 - Small, well defined synthesis IC2 - Ideating initial solutions RU1 - Targeted refactoring RU2 - Shallow understanding VD1 - Validation needs human VD2 - Debugging is context sensitive								
Respondent ID	Participant Programming Experience	Participant No.	Responses	Short-Comments	Labels	# IC1	# IC2	# RU-1	# RU-2	# VD-1	# VD-2
110019533756		24	P1 I'm not sure this is the right way to categorize tasks. ChatGPT is useful if I'm doing something that has been done thousands of times. It's useless if I'm doing something that has never been done before.	Good on common/repeated tasks; poor on novel/OOD problems.	IC-1	1	0	0	0	0	0
110019511082		P2	Every time I've used ChatGPT to debug it's gotten the answer wrong multiple times and sometimes given impossible or illogical answers with complete confidence (including ones that don't even type check or compile). I do not trust it to debug nor validate security etc because of how prone it is to hallucinations. And I am always much faster on my own (it is a net negative).	Worst at debug/validation (hallucinations); understanding mid but unreliable; brainstorming generic; shines at codegen/refactor; validation loops add cost.	IC-1, IC-2, RU-2, RU-1, VD-1, VD-2						
		13	Understanding is somewhere in the middle; LLMs are good at summarizing but they can also hallucinate and incorrectly explain things, making it very unreliable and untrustworthy. ChatGPT in particular is horrendous at even trying to cite its answers. I hear Perplexity is better. I'd much much rather read from the source in nearly every case (for older and less documented codebases though, it can be useful to summarize some chunks)			1	1	1	1	1	1
			It can give a quick write-up for brainstorming, but it only gives common ideas for the most part and for novel solutions it will hallucinate and so is not reliable. I don't really need assistance in thinking of common solutions. For novel or newer architectures (e.g. lockless, coordination-free, etc), it can be useful to bounce off another significantly experienced developer to find flaws in the architecture. ChatGPT's glazing, hallucinations, and common patterns unfortunately make it not very helpful in such a process (where it could otherwise be useful).								
			Refactoring and code generation are where it shines better given the output to time ratio. But good automation and boilerplate goes a long way already. And of course, it can give outdated boilerplate and refactor incorrectly or with poor style. But if you have it go through a validation loop (i.e. check its work against type checker, compiler, tests, etc), it can be more accurate. But that reduces its efficiency and requires more prompting, so may not be worth the increased effort depending on the size of the codebase								
110019505086		5	P3 When it comes to new ideas AI tools are least efficient. I find it more useful to use AI for tedious tasks such as code generation, debugging or refactoring. These task although could be done manually, I consider them low skill since AI has emerged.	Ideation least efficient; good for tedious codegen/debug/refactor.	IC-1, RU-1	1	0	1	0	0	0
110019432168		9	P6 Generating new code based on billions of code snippets seems to work well for smaller tasks. The same is true for debugging, when it builds its answers from StackOverflow threads and so on that we would have to search manually. For more complex code incorporating several files and modules, it often feels like its getting confused.	Codegen/debug OK for small tasks; confused on multi-file complexity.	IC-1, VD-2	1	0	0	0	0	1
110019427134		19	P7 For me it is still a language model that is good at summarizing and synthesizing. The further away from that, the worse.	Strong at summarizing/synthesizing; worse beyond that.	IC-1, RU-2	1	0	0	1	0	0
110019425850		7	P8 generating abstract ideas is easier to do with AI tools. However, when there are specific and numerous parameters effect the context, then AI tools become less effective, atleast from my experience. So, the easiness of debugging depends on the context, however, in many cases, code do have multiple contexts, which needed to be explained further.	Abstract ideas easy; effectiveness drops with many parameters; debugging highly context-dependent.	IC-2, VD-2	0	1	0	0	0	1
110019425258		15	P9 I don't find chat GPT useful except for explaining things that are hard to google for	Mainly useful for explaining things hard to Google.	IC-2	0	1	0	0	0	0
110019424217		10	P10 Because I use ChatGPT in such way			0	0	0	0	0	0
110019423107		25	P12 It's good for one of two statements, but not a complex logic that spans multiple pages/methods.	Fine for short/simple snippets; not for complex multi-page logic.	IC-1	1	0	0	0	0	0
110019411831		10	P13 It is great to translate what you write into code or to be a rubber duck. Have a hard time with debugging, often suggesting wrong fixes, or easy fixes that have already been tried.	Good as rubber-duck/translate to code; poor debugging (wrong/easy fixes).	IC-1, VD-2	1	0	0	0	0	1
110019408832		11	P14 It is a fairly decent tool at spitting out chunks of code, but when it comes to any kind of nuanced understanding of code or coming up with novel ideas, it fails miserably. Even when it generates code, it most often requires the eye of an experienced developer to validate that its solution doesn't introduce unwanted corner cases.	Decent chunks of code; weak nuanced understanding/novel ideas; human validation needed.	IC-1, RU-2, VD-1	1	0	0	1	1	0
110019407538		13	P15 ChatGPT can be useful for inspiration and sometimes for understanding. But everything else it's pretty weak at, because ultimately it does not truly understand the code. I maintain a large repo on github and AI-generated PRs are a problem we're facing.	Inspiration/occasional understanding; lacks true code understanding; AI PRs problematic.	IC-2, RU-2	0	1	0	1	0	0
110019406801		15	P16 I find it pretty useful in Understanding when it comes to projects which involve many technologies I'm not familiar with. It's the same reason for Debugging is least effective, as ChatGPT can't identify the exact problem when there are too many technologies involved.	Helpful for understanding unfamiliar, multi-tech projects; debugging worst when many techs involved.	IC-2, RU-2, VD-2	0	1	0	1	0	1
110019399151		P17	It doesn't understand anything, it just vomits slop, and it is confidently wrong.	Often confidently wrong; good for exploring options and boilerplate/unit-test scaffolds.	IC-2, IC-1, VD-1						
		24	brainstorming is high, because it's great at exploring options you may not know about.			1	1	0	0	1	0
			code generation because it saves you from writing boilerplate, and it's okay at doing unit tesfs if you have a good description of the test								
110019397914		27	P19 Debugging and refactoring often require a lot more context, also non-code context and impacts a lot of files at once. Increasing chances of mistakes or misunderstandings.	Debug/refactor need lots of non-code & cross-file context → more mistakes.	VD-2, RU-2	0	0	0	1	0	1
110019397543		12	P20 it kinda doesn't understand a lot of optimizations, including declaring variables used inside a loop outside of the loop. It sometimes insists it's a bug since the values are never used outside of the loop. Really the only good thing it's good for is documenting code, and even then it only understands what the code does if you tell it, and at that point you might as well document it yourself.	Misses optimizations; OK for documentation if you supply meaning.	RU-2	0	0	0	1	0	0
110019397532		15	P21 ChatGPT is better at open-ended tasks that don't have a "right answer"	Better for open-ended tasks without a single right answer.	IC-2	0	1	0	0	0	0
110019395380		11	P22 I typically use Claude Sonnet via Cursor, but honestly it was hard to rank because I've used it for all of those things and it's incredibly impressive at all of them. I would trust it with any of those tasks as a first step.	Impressive across tasks; reasonable as a first step.	IC-2	0	1	0	0	0	0
110019396235		14	P23 Works best with well defined tasks like refactoring or basic code generation. Difficult to effectively manage context for more open ended tasks and the feedback loop latency is high.	Best for well-defined refactors/basic generation; struggles with open-ended context.	IC-1, RU-1	1	0	1	0	0	0

110019395936	6	P24	I do not find ChatGPT effective for any of these tasks but had to mark them to move forward in the survey.			0	0	0	0	0	0
110019395348	28	P25	never used it for validation. most of the time it creates the boilerplate and I start from there or ask it for idea / code investigation.	Rarely use for validation; often start from generated boilerplate.	IC-1	1	0	0	0	0	0
110019395345	10	P26	Chat GPT makes devs (and myself) understand their codebases less. Your brain on gpt made an impact on me and others	Harms deep understanding of one's own codebase.	RU-2	0	0	0	1	0	0
110019394986	20	P28	Brainstorming is absolute top. It really helps with decision - it's lengthy, it requires explaining context, knowledge of domain, but it can give pros and cons. It's good with debugging to point in what direction to look. It's not useful with helping, but with patience you can go through process of removing causes one after other until cause is found. Again, as much of context needs to be given to the LLM. It can explain code really well, step by step or section by section. I do use checks for logic errors like writing wrong variable name, or forgetting something during refactoring. ChatGPT is good to keep connections if you provide code. However it's absolutely terrible at proposing new code which try to use interconnected code across various objects. This also is true with refactoring. It loses context pretty quickly and tends to hallucinate "its own code" even if file is directly provided. Which leaves code generation - for short or framework you build upon it's good. If you want it to extend your code, then it's bad. Introduces logical errors, leaves out parts of code and for some reason it cannot address line numbers to show diff, or worse, it shows line of code which is similar in functionality, but it's not what's written. So for code generation - the framework, ok, but you finish it and connect it to other components.	Brainstorming top; debug gives direction with context; explains code well; large refactors/context → hallucinations; short/framework codegen OK.	IC-3, IC-2, RU-1, VD-2, IC-1						
110019395125		P29	It's best at generating code in situations where the code has been commonly written.	Best when generating commonly written code.	IC-1	1	0	0	0	0	0
110019395016		P30	(this goes beyond ChatGPT - I use many AI tools, including OpenAI Codex, Cursor, Claude Code, etc.) Current AI coding tools still need gut checked - they make mistakes and hallucinate from time-to-time. It's getting less and less frequent, but the fact that I can't trust the tool 100% yet means I can't use it extensively for validation. Debugging error codes... it's ok at sometimes, but probably about as effective as me googling for solutions. It's very good at planning (brainstorming/ideation) and code generation, but I *always* have to double check its work and ask it to fix mistakes. It saves me loads of time - but not so much that I can completely hand tasks off to it.	Needs gut-check; validation not trusted; brainstorming/codegen helpful but must double-check.	VD-1, VD-2, IC-1, IC-2	1	1	0	0	1	1
110019390040		P31	It can quickly spit out lots and lots of code which mostly works, but it is not expected to have an understanding of any actual issues.	Can emit lots of mostly-working code; lacks grasp of underlying issues.	IC-1, RU-2	1	0	0	1	0	0
110019383854	10	P32	In my experience, tools like ChatGPT and others are invaluable for navigating the initial challenges of engaging with a topic, enabling meaningful interaction without necessitating a foundational skill level. However, they cannot substitute for authentic insight and real-world experience. Complex problems that require a more profound understanding—those that appear infrequently in any dataset—often demand extensive guidance, repeated iterations, and support far beyond what it would take to solve them independently. This isn't surprising, given that educational resources in coding tend to focus on a lower skill ceiling. The steep learning curve means that aspiring developers who never attain a sufficient skill level outnumber those who do, all while the job market for developers continues to expand rapidly. As a result, the field is flooded with amateur talent. These dynamics result in a significant overrepresentation of sample data from individuals with below-average abilities, even when measured against the already skewed skill curve inherent in technical subjects. Consequently, AI can often be likened to a fast but relatively inexperienced junior developer. TL;DR: AI (of the current approach, at least) are forever stunted thanks to the chronic skill issue they were born with. But they can potentially make for great librarians	Great for initial engagement; can't replace deep expertise; complex/rare problems need heavy guidance.	IC-2, RU-2	0	1	0	1	0	0
110019382325		P33	ChatGPT tends to give me a sense of direction when I'm trying to solve a new problem or when I'm implementing a new feature. I can't justify it but I have mistrust in using it for refactoring, optimization or for bug hunting.	Good for initial direction; mistrust for refactor/optimization/bug-hunting.	IC-2, VD-1, RU-1	0	1	1	0	1	0
110019382266		P34	It is good in areas where you already have code or an idea and it is not necessary to generate a lot of code (because it is very unreliable for that).	OK when you already have code/idea; unreliable for large-scale generation.	IC-1	1	0	0	0	0	0
110019380610		P35	Validation requires a brain. ChatGPT is a glorified probability matrix (logits->softmax). Can not comprehend even basic thought due to it's deficient design, it can only spit out probabilities that were burned into it's weights, usually verbatim. Hence it's ability to spit any idea/ideology verbatim, all at the same time, when you prompt it a particular way. For everything else I can act as the brain. It's a dumb tool, that is all. A great tool. But a dumb one.	Validation requires human judgment; tool lacks genuine comprehension.	VD-1	0	0	0	0	1	0
110019378988	20	P36	ChatGPT can easily match code with similar patterns, making it great for refactoring into another pattern, or recognizing common patterns for generating an explanation. That's why refactoring and understanding are my top two. ChatGPT's brainstorming is great at generating good-enough, 6/10 solutions. But the best ideas on our team always come from a protracted, iterative, human cycle. ChatGPT can facilitate that human interaction, but it's rarely the source of the ideas. ChatGPT's sycophancy means that I've had trouble with it validating my code. It just says "yeah that works the way you expect it to", making the same mistakes as I did in the process	Pattern matching helps refactor/explain; brainstorming gives "good-enough" options; sycophancy harms validation.	RU-1, IC-2, VD-1	0	1	1	0	1	0
110019378481	12	P37	ChatGPT is better at parsing large pieces of code and transforming them in some way, like for debugging or refactoring or generation	Good at parsing/transforming larger code chunks (debug/refactor/gen).	RU-1, IC-1	1	0	1	0	0	0
110019377725	8	P38	too opinionated, biased			0	0	0	0	0	0

110019376658		P40	ChatGPT is so good to brainstorming, as the interace is conversational, after the ocnversation is possible to ask to chatGPT to generate some documentations. The models for debugging are really good too. For complex logic in code the conversational interface helps a lot too. The refactoring, code generation and validation, in my case as a staff SW, are less important, but helpfull to improve the development performance.	Strong at brainstorming/conversational docs; mixed views on others.	IC-2, RU-2	0	1	0	1	0	0
	12										
110019375841		P41	I do not find chatgpt helpful basically for any of these tasks. I would have ranked all of them at 5-6 if I could. Chatgpt is pedantic, annoying, and unpleasant to work with.			0	0	0	0	0	0
110019375009		P42	essentially ranked from least exacting to most exacting. Basically the consequence of a ChatGPT error goes dramatically up, throughout that ordering			0	0	0	0	0	0
110019374848		P43	It's effective at teaching me unfamiliar topics, but it hallucinates for "out of distribution" tasks (obscure or idiosyncratic topics)	Teaches unfamiliar topics; hallucinates on OOD/obscure cases.	IC-2	0	1	0	0	0	0
110019374411		P44	Specifically ChatGPT is useless for all these tasks except occasionally as a search engine for something you don't know what keywords to look for and generating very small code snippets (which normally need rewriting anyway but at least demonstrate a technique). Claude is quite good at very targeted refactors ("update this method signature and all usages of it". Never something open ended like "make my code neater/faster") or tasks where there's only really one possible intended output. If you give it something open ended like asking it to write tests or documentation then it comes up with something that's believable at a glance but also rubbish.	Mostly useless; except search/small snippets; targeted refactors OK; open-ended outputs poor.	IC-1, RU-1, RU-2	1	0	1	1	0	0
	8										
110019374398		P45	I roughly ordered them for tolerance to error.			0	0	0	0	0	0
110019374190		P46	Generative AI is just advanced pattern recognition i.e. "more accurate text prediction". It might be useful in situations where basic prediction is helpful, for inexperienced junior developers etc. But the kind of problems I solve require actual understanding and not merely regurgitating patterns.	Pattern recognition helps basics; lacks real understanding needed for my tasks.	RU-2	0	0	0	1	0	0
	40										
110019374153		P47	ChatGPT (Copilot) analyses multiple sources and summarise from multiple sources much faster than I can on my own, and will frequently mix in my own context from previous searches, to produce high quality suggestions. It enables me to iterate my analysis very quickly to learn what I need.	Multi-source summarization + personal context → fast analysis.	IC-2	0	1	0	0	0	0
	43										
110019374111		P48	I don't trust ChatGPT to be always correct, therefore it is better suited for tasks that don't need a very high confidence.	Use for low-confidence tasks; don't trust full correctness.	VD-1	0	0	0	0	1	0
	20										
110019374012		P49	ChatGPT is very good at explaining code errors.	Very good at explaining code errors.	RU-2	0	0	0	1	0	0
110019373880		P51	ChatGPT struggles with original ideas, which are common in open source, which makes it bad at understanding and validation. It also struggles with managing all the constraints needed for any meaningful refactor.	Weak on original ideas/validation; struggles with constraints in real refactors.	IC-2, VD-1, RU-1	0	1	1	0	1	0
110019373745		P52	I use chatgpt to understand what code does, and suggest ideas. In my experience, it doesn't reliably generate working code without a lot of to-and-fro and human intervention, but I still find conversing with chatgpt to be helpful in understanding the problem domain. Sometimes the back-and-forth helps me to understand the problem better and gives me an idea for a different solution, having reached dead ends with chatgpt's suggestions.	Use to understand code and spark ideas; codegen unreliable without back-and-forth; good at "why error?"	IC-1, IC-2, RU-2	1	1	0	1	0	0
	30		I have had success asking why this code gives this error, and I rely on this a lot.								
110019373698		P53	I use AI for most of these topics- it's especially useful for generating "boilerplate" code in a language where I lack muscle memory.	Great for boilerplate; brainstorming conventional approaches; validation is people-pleasing.	IC-1, IC-2, VD-1						
			Brainstorming with AI is most useful for understanding the "conventional" approach to a topic, which is very helpful, especially when I'm low context in the area (it's like an excellent generalist)			1	1	0	0	1	0
	19		Validation is the only topic where I've found AI to struggle; it can be kind of "people pleasing" in a review, where it tends to want to compliment the solution, and can miss opportunities for improvement.								
110019373505		P55	ChatGPT, Cursor, Copilot, Perplexity, et al. can be useful for generating and refactoring code, but overall are far less suitable for everyday tasks than someone who knows what they're doing.	Useful for gen/refactor but weaker than an expert overall.	IC-1, RU-1	1	0	1	0	0	0
110019372009		P56	In my experience, I've seen that chatgpt understands the context of a program for which it can figure out many ways or steps for issue resolution. Also, For simple method level code generation, chatgpt performs quite well. However, for refactoring a medium to big sized program, chatgpt struggles to figure out the best coding practices for that program.	Understands context to outline steps; method-level gen OK; medium/large refactors struggle.	IC-1, RU-1	1	0	1	0	0	0
	7										
110019370799		P57	Sometimes it's useful to use copilot to start out a snippet of code, and then refactor it to make the code more efficient or more readable/maintainable. In frameworks or languages I am less familiar with, sometimes I ask GPT if there is a well-known way to make my code more efficient ("Refactoring"). I do not trust that ChatGPT has a deep understanding of the code it generates, at least not an understanding that is comparable to human understanding, which is why I ranked "Understanding" last.	Start with snippet then refactor; asks for efficiency tips; distrusts "understanding."	IC-1, RU-1, RU-2	1	0	1	1	0	0
	10										
110019370134		P58	ChatGPT good at giving initial idea	Good for initial ideas.	IC-2	0	1	0	0	0	0
110019367335		P59	Ideation and understanding are two of the most important/effective given the model's broad knowledge. Refactoring can be trickier for the models, often requiring lots of steering	Ideation/understanding effective; refactoring needs heavy steering.	IC-2, RU-2, RU-1	0	1	1	1	0	0
110019367121		P60	Having crawled the whole open web, I find ChatGPT very useful to point you toward the right tools to do a task (brainstorming). However, its statistical approach to text generation make it poorly suitable to really develop a reasoning about bug and large project refactoring.	Good at pointing to right tools; weak on bugs & large refactors.	IC-2, RU-1	0	1	1	0	0	0
110019365016		P61	ChatGPT helps me understand unfamiliar technology. I had to understand Vector and VRL language, InfluxDB and Flux queries, and kubernetes. All of these topics were very unfamiliar to me. ChatGPT helped me "get me feet wet" with the technology so I could build vector pipelines and flux queries. However, I ended up purchasing books from Manning publications to further cement my understanding of this tech.	Helped ramp on unfamiliar tech; codegen OK but don't paste blindly.	IC-2	0	1	0	0	0	0
	8		ChatGPT is ok with code generation, but I do not blindly copy and paste. I understand businesses want us to be productive and to produce code as fast as possible, but I am a human and I just want to have fun while I still can.								
110019362616		P62	I have never used ChatGPT myself. I consider it to be the first resort of incompetent software developers. What such people want to do, in my opinion, is to have an artificial intelligence process generate code for them, which can then be applied to the person's current issue or overall project and then passed off as his or her own code. Such a person doesn't feel a need to understand the specific programming language or the business logic that is needed. ChatGPT works best as a brainstorming tool, to help give a workgroup useful ideas in resolving the issue at hand. At this time, I think ChatGPT (and some other AI engines) simply cannot substitute for a team of developers willing to do careful code reviews.	Personally don't use; sees value mainly in brainstorming.	IC-2	0	1	0	0	0	0

110019363194	25	P63	I use copilot, not ChatGPT. Answering assuming you're asking about copilot. Copilot often generates too much code. Hand crafted code is much more succinct. So I don't usually request it to write lots of code for me at once. It's most useful when I am working in an unfamiliar language. I can ask it how to create a particular idea when I don't know the syntax. It's not great at test writing since it is too verbose, making the tests hard to read. In particular, it tends to add lots and lots of mocks even when calling the base code directly is sufficient for the test.	Too much code; good for unfamiliar language/syntax; weak at tests (verbose/mocks).	IC-1, IC-2	1	1	0	0	0	0
110019363180	10	P64	1. GPT can generate most of the common snippets and writing them in a extremely friendly way that human developers normally don't bother to do. 2. ChatGPT can refer to a lot of documentation, so it can help developers understand the code very quickly. 3. It can give a lot of insight, but some of them might be wrong. 4. It can solve a lot of problems only if they're something you can find in the docs or online forums. 5. It can only help you to do very simple refactoring that not worthy to ask. 6. It couldn't really understand the logic and the correctness of the code.	Common snippets good; fast understanding from docs; only simple refactors; lacks logic/correctness.	IC-1, IC-2, RU-1, RU-2	1	1	1	1	0	0
110019362923	20	P65	I like mostly to learn from chatgpt, and use it initially. Solving the issue is more my brains work.	Use to learn initially; solve with own reasoning.	IC-2	0	1	0	0	0	0
110019360136	15	P66	I find that ChatGPT is really good at brainstorming, decent at putting out "mostly correct". I find it's not great at understanding foreign code. Shrug.	Good brainstorming; "mostly correct" outputs; weak at understanding foreign code.	IC-2, RU-2	0	1	0	1	0	0
110019360067	45	P67	I don't use it. They are all least effective, but the survey forced me to put in numbers.			0	0	0	0	0	0
110019359284	30	P68	I've tried to use ChatGPT for all these things and it basically failed completely. I tested ChatGPT on bugs that I tracked down myself. It completely failed at identifying the bugs and just gave generic suggestions. Then I tested it on the fixes and it was almost as bad, giving useless or harmful suggestions. The only thing I have found it to be useful for is to get some idea of what functions from a library or SDK I might use to solve some problem. I'll ask it to generate some code, look at what functions it used, and then go and look up those functions in the real docs and implement the code myself. The code generated by ChatGPT probably doesn't work but usually at least the function names are right.	Failed at bugs/fixes; only useful for surfacing candidate APIs to check.	VD-1, IC-2	0	1	0	0	1	0
110019352947	7	P71	I noticed most of the time chatgpt agrees with what user says even though it is not true.	Tends to agree with user even when wrong (sycophancy).	VD-1	0	0	0	0	1	0
110019352649	5	P72	To generate ideas and to understand the issue, I can brainstorm with ChatGPT. It helps me to get the explanation fully.	Brainstorm to generate ideas and understand issues; explanations help.	IC-2, RU-2	0	1	0	1	0	0
						25	30	15	20	12	8